

Scientific Programming with Python

Assignment: The Interaction Between Two Atoms

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Goal The goal of this assignment is to make use of the knowledge about scientific computing learned so far. Specifically, you will learn about the Lennard-Jones [1] potential for modeling molecular interactions, unit conversion, implement user-defined functions and explore the Pandas library [2, 3].

Problem and Input Data The simplest classical-physics model for how atoms interact is the Lennard-Jones equation [1], which has the following form:

$$V_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (1)$$

In the equation, $V_{LJ}(r)$ is the potential energy of the interaction, ϵ (energy well depth) and σ (distance when $V_{LJ}(r) = 0$) terms are atom-pair dependent parameters, and r is the actual distance between the atoms. Michels et al. published the following suggestions for ϵ and σ parameters for Argon (Ar), which are given in Table 1 [4].

Table 1. Argon parameters for use in Lennard-Jones (Eq., 1) equation.
The original units were given in ergs and cm [4], which have been converted to the more modern units.

Species	ϵ (kJ/mol)	σ (Å)
Ar	0.995	3.405

Finally, Cybulski and Toczyłowski used very accurate quantum mechanics (QM) calculations to determine $V(r)$ (in Hartree) at different r distances for several rare gas dimers – including Ar_2 – which can be used as benchmark target data [5]. The data from these calculations can be found in the `CybulskiT1999_Ar2.csv` file.

In this assignment you will encode the Lennard-Jones data and computer the potential energy of Ar_2 at several distances using existing ϵ and σ literature values. The results will then be compared to QM data and visualized.

Assignment Tasks

Task 1 Load and prepare the QM target data into a Pandas' DataFrame.

- Load the data for Ar_2 found in `CybulskiT1999_Ar2.csv`.
- Clean this data by removing rows with duplicated or missing data.
- Convert the potential energies from Hartree to kJ/mol units.

Task 2 Compute the Lennard-Jones potential energy (Eq. 1) using the distances found in the QM target data and the parameters given in Table 1. Hint: you should consider here the idea of user-defined functions.

Task 3 Visualize the data by plotting the following using lines:

- the QM target energies (i.e., kJ/mol units), and
- the Lennard-Jones energies (i.e., computed in Task 2)

Hint: The x-axis should be the distances (i.e., R (Å)).

Task 4 Save the following data to a CSV-formatted file:

- the distances (Å)
- the QM target energies in Hartree,
- the QM target energies in kJ/mol, and
- the Lennard-Jones energies (kJ/mol)

Allowed Python3 [6, 7] functions & libraries/modules

- All built-in functions
- Pandas library [2, 3]
- math library
- typing library (if needed)

Assignment Due Turn in your solution as a **Jupyter-notebook** [8] to **LEA** by **Tuesday, May 19th, 2026 at 13:30**.

Note 1 : Please include your **SciPro.ID** at the **top** of your notebook.

Note 2 : **Do not** consider **significant figures** in this solution.

References

- [1] Wikipedia contributors, "Lennard-Jones potential." Wikimedia Foundation. Last modified July 24, 2022. https://en.wikipedia.org/wiki/Lennard-Jones_potential. Accessed on September 12, 2023.
- [2] The Pandas Development Team pandas-dev/pandas: Pandas Zenodo, 2020 (<https://pandas.pydata.org>)
- [3] Pandas user guide. Available at https://pandas.pydata.org/docs/user_guide/index.html. Accessed on November 7, 2023.
- [4] Michels, A., Wijker, Hub., & Wijker, Hk. (1949). Isotherms of Argon Between 0°C and 150°C and Pressures Up to 2900 Atmospheres. *Physica*, 15(7), 627-633.
- [5] Cybulski, S.M. & Toczyłowski, R.R. (1999) Ground state potential energy curves for He₂, Ne₂, Ar₂, He-Ne, He-Ar, and Ne-Ar: A Coupled-Cluster Study *J. Chem. Phys.*, 111, 10520-10528
- [6] Python Software Foundation. Python Language Reference, version 3.11. Available at <http://www.python.org>. Accessed on October 22, 2023.
- [7] van Rossum, G. Python tutorial, Technical Report CS-R9526, Centrum voor Wiskunde en Informatica (CWI), Amsterdam, 1995.
- [8] Kluyver, T. et al., (2016) Jupyter Notebooks – a publishing format for reproducible computational workflows. In F. Loizides & B. Schmidt, eds. *Positioning and Power in Academic Publishing: Players, Agents and Agendas*. pp. 87-90.